Automatic Ticket Classification with Flow Designer

Problem Statement

Every day the school IT helpdesk is flooded with incident requests from students and teachers, covering things like Wi-Fi outages, broken projectors, forgotten passwords and sluggish computers. Right now a member of the IT staff has to read each request and choose its category by hand, which is slow and makes poor use of their time.

This project takes that job away from people. Using ServiceNow Flow Designer, it scans the short description and description of each incident for keywords and classifies the ticket automatically.

Objective

- Classify each incident automatically at the moment it is created.
- Lower the amount of manual work that IT agents have to do.
- Get tickets to the right destination faster.
- Provide a no-code solution that is simple to maintain.

PHASE 1: Requirement Analysis & Planning

Students and teachers send the school helpdesk requests every day about familiar problems: network connectivity, hardware faults, account access and slow system performance. At present, IT staff open every ticket themselves so that they can:

- Work out what kind of issue it is.
- Choose the correct category and subcategory.
- Tell the caller that their ticket has been logged.

Working this way consumes a lot of time, invites mistakes and will not cope as the number of requests grows.

Activity 1: Business Use Case

Business Requirements

The system needs to:

1. Classify IT tickets automatically from the description of the issue.
2. Fill in both Category and Subcategory without anyone stepping in.
3. Offer dependent choice behaviour between Category and Subcategory.
4. Email the caller automatically as soon as a ticket is created.
5. Keep ticket data in a structured, standard format.

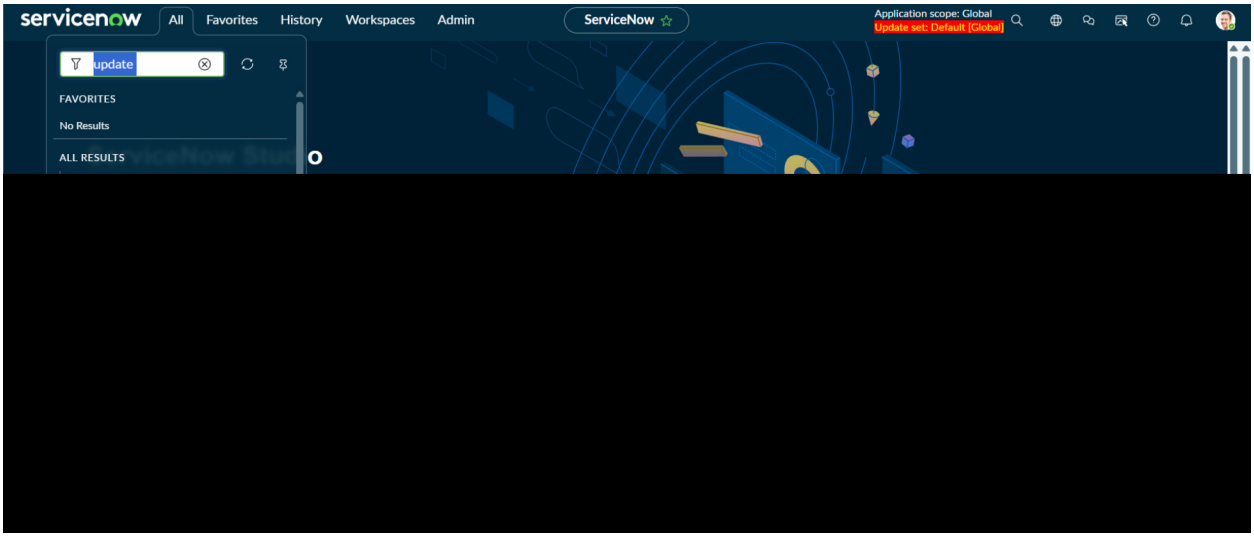
6. Remain easy to maintain and ready to grow in the future.

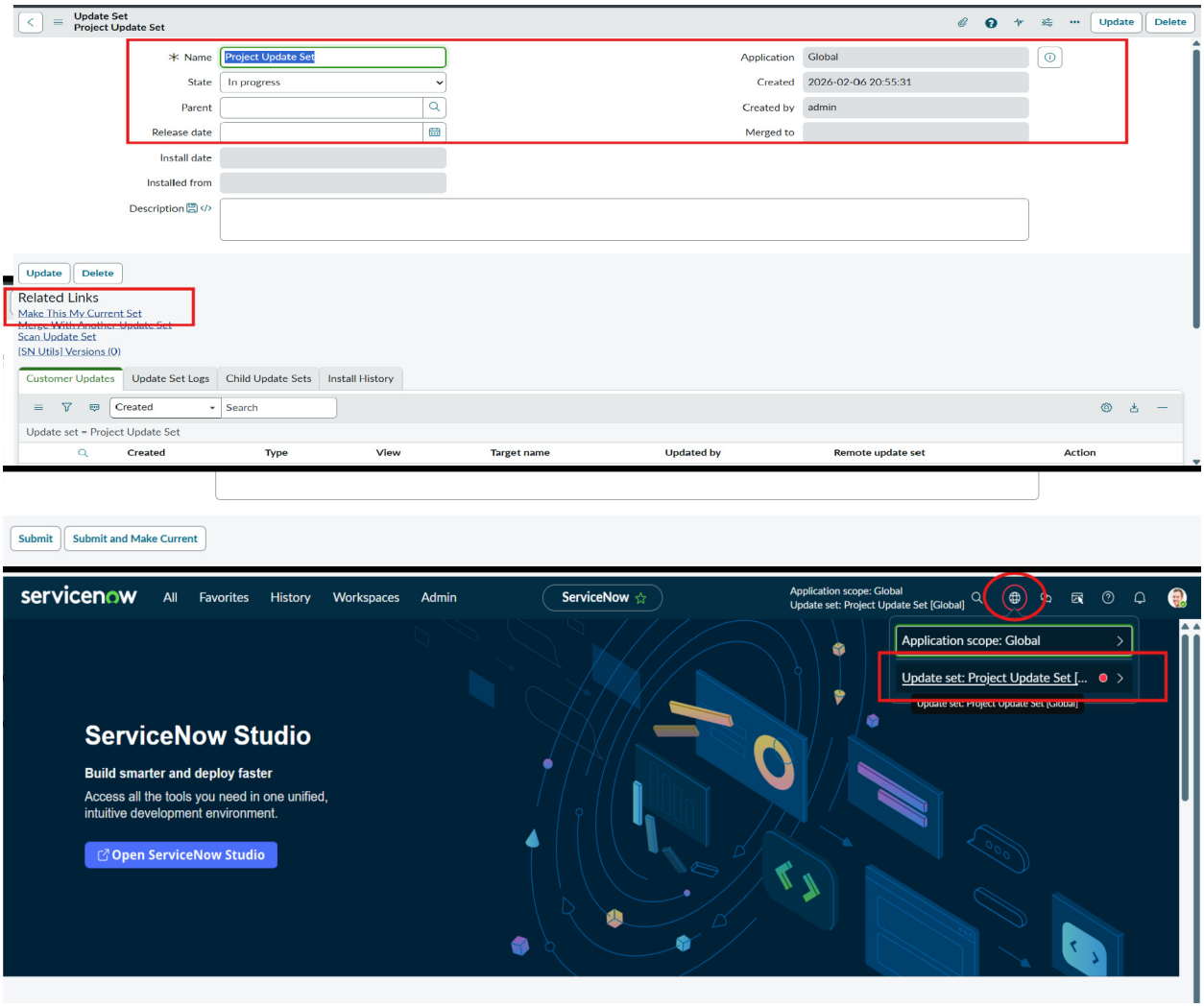
Activity 2: Creating a New Update Set

Navigation Steps

- Open the All menu and search for Update Set.
- Under System Update Sets, choose Local Update Sets.
- Press the **New** button at the top right of the page.
- Enter the name **Project Update Set**.
- Set State to **In progress**.
- Set Application to **Global**.
- Save the form.
- In Related Links, click **Make This My Current Set**.
- The new update set now shows up as the current one.

Activity 3: Navigation Steps in Picture View



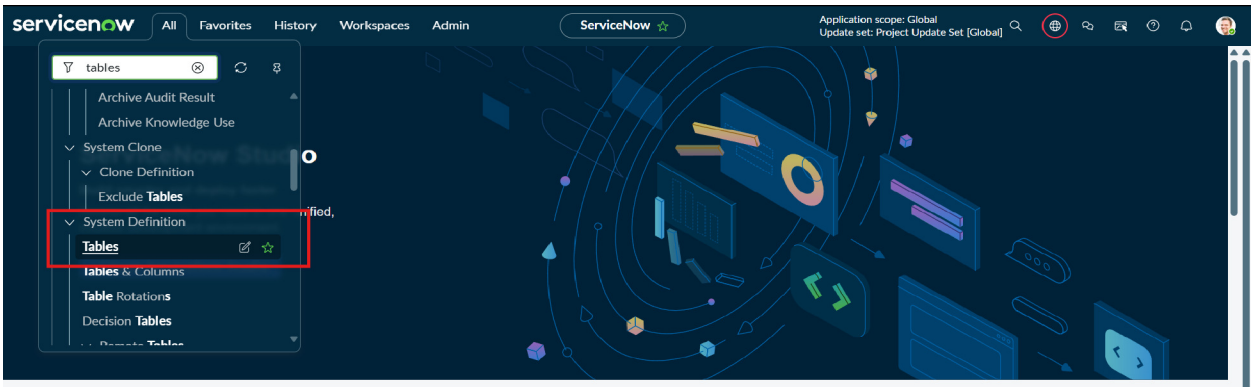


PHASE 2: Backend Development & Configuration

Activity 1: Creating a Custom Table for Ticket Records

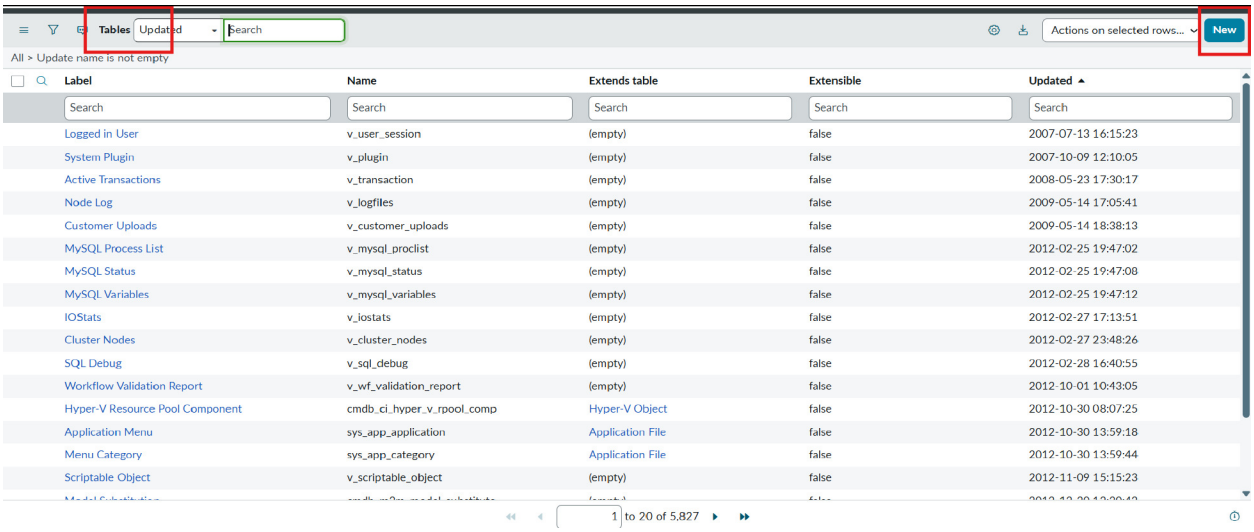
Steps to Create the Custom Table

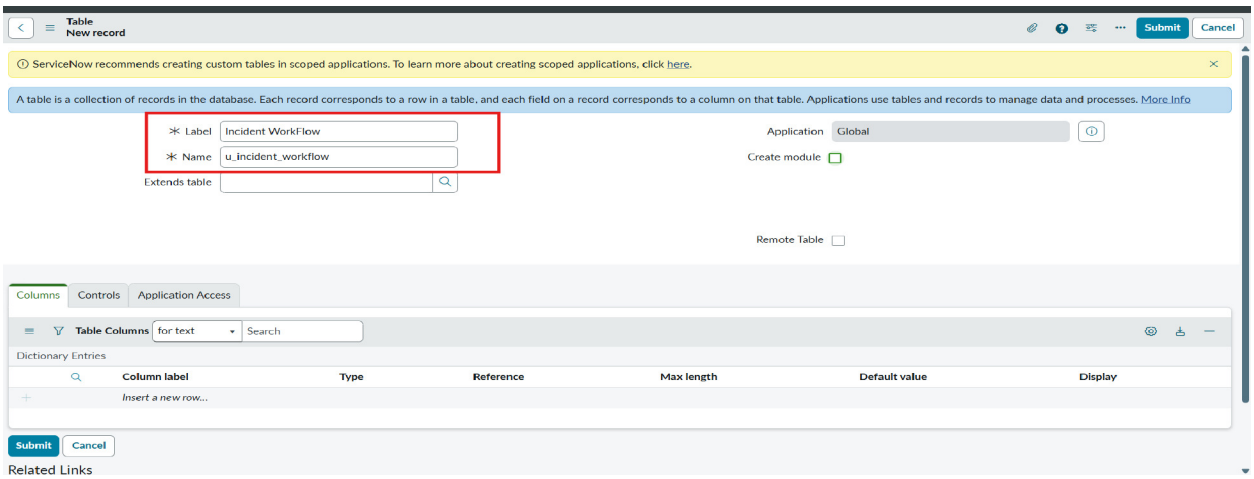
- Open the All menu and search for Tables.
- Under System Definition, click **Tables**.
- Click the **New** button at the top right.
- Give it the label **Incident WorkFlow**.
- Untick **Create module**.
- Save the form.
- Go to the **Controls** tab and tick the **Auto Number** checkbox.
- Save the form again.

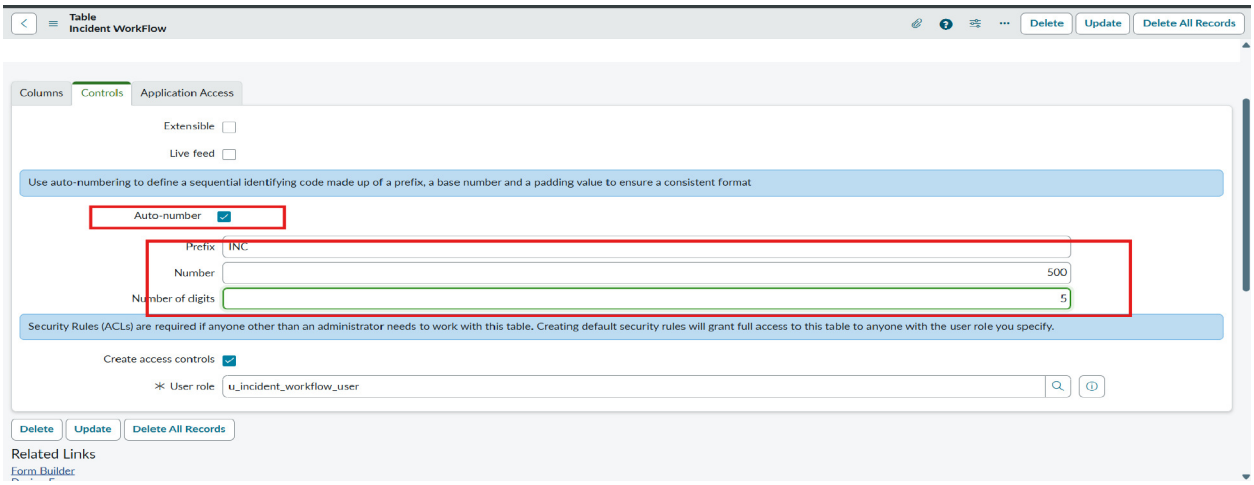


GO FURTHER

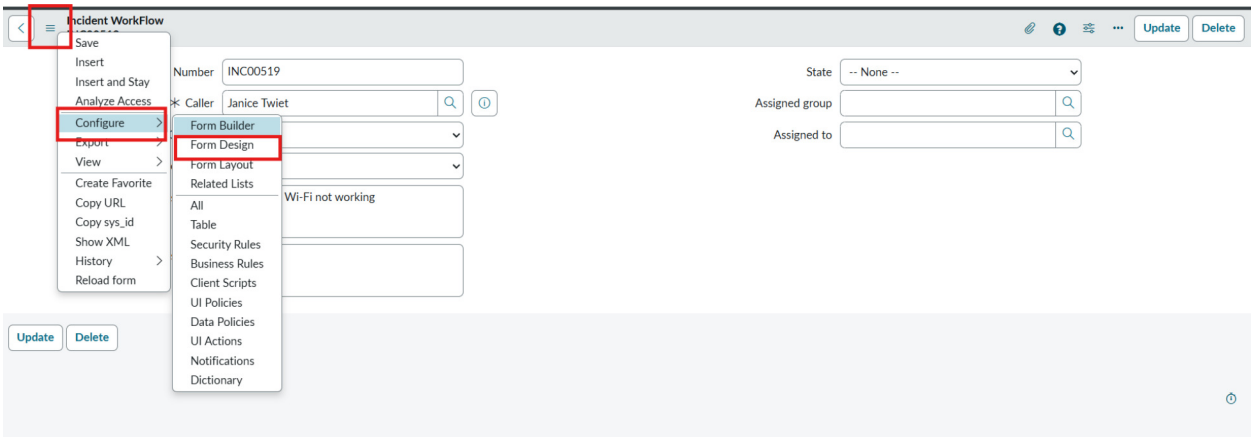
Power your workflow applications







- After saving, click **Show Form** under Related Links.
- Open the context menu, go to Configure and choose **Form Designer**.
- Add whichever fields the table needs.

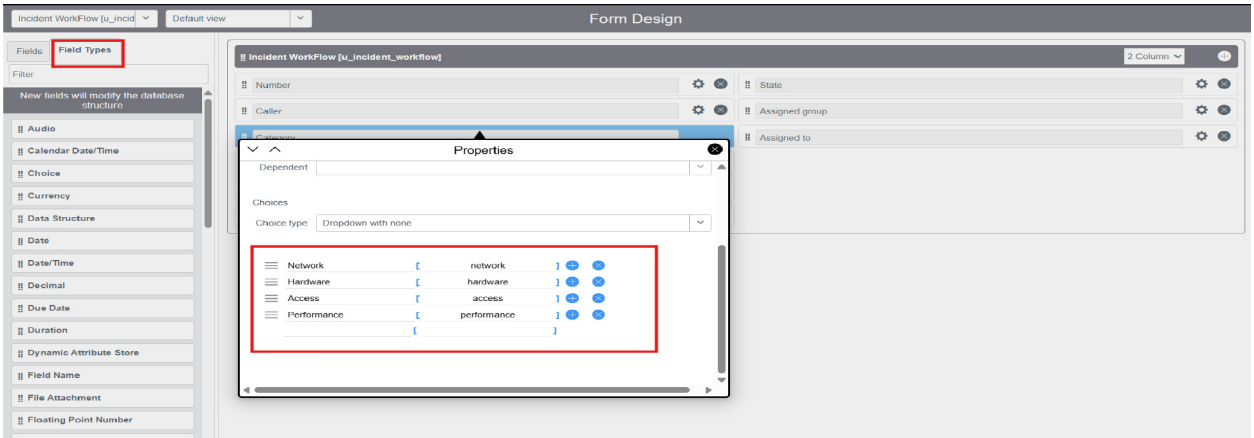


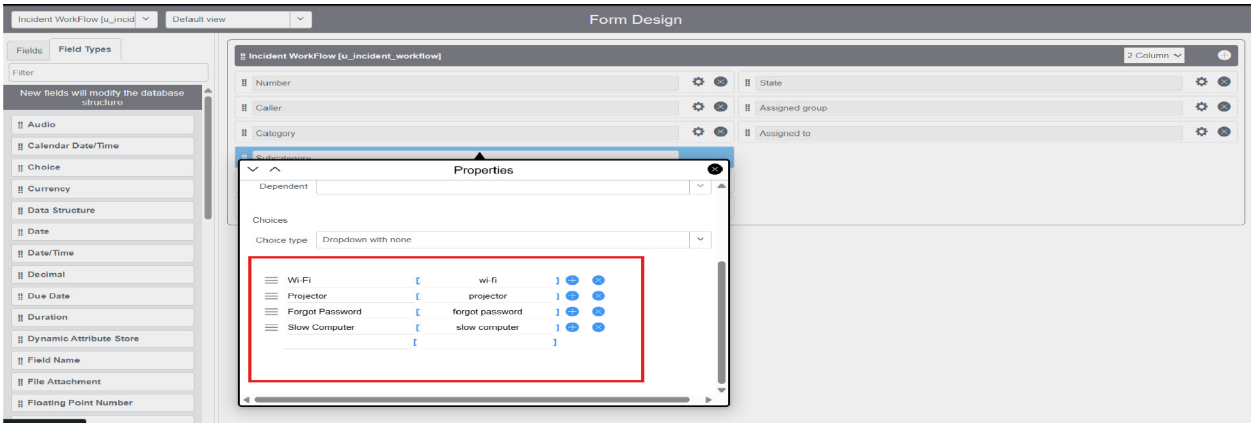
Activity 2: Creating Fields and Setting Data Types

This milestone builds every field that the School IT Ticket table requires, each with the right data type, choice values and dependencies, so that the data captured is structured and consistent.

Field Label	Type	Reference / Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network, Hardware, Access, Performance]
Subcategory	Choice	[Wi-Fi, Projector, Forgot Password, Slow Computer]
Short Description	String	–
Description	String	–
State	Choice	[New, In progress, On hold, Resolved, Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- Once the fields exist with their proper types, define the choice values from Form Designer.
- Do this for every choice field: Category, Subcategory and State.





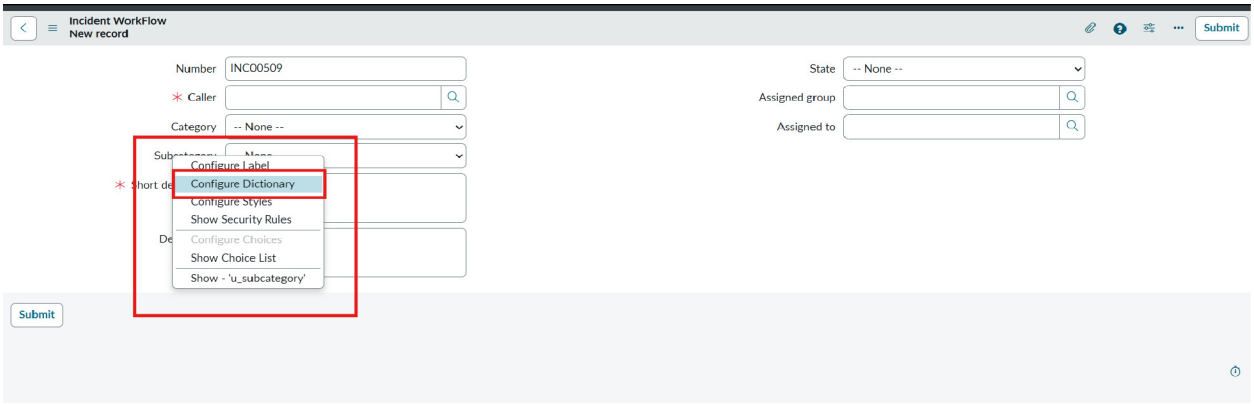
- Add the State choices in the same way.
- Click Save at the top right.
- Open the form and refresh it.

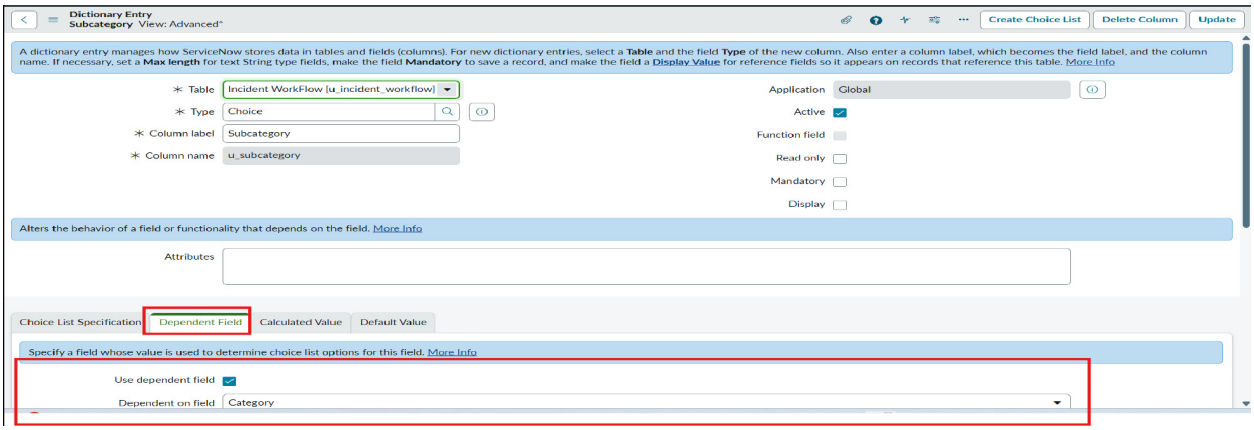
Activity 3: Setting Up the Dependency Between Category and Subcategory

The goal of this milestone is to tie the Category and Subcategory fields together, so that the Subcategory options change on their own according to the Category that has been selected.

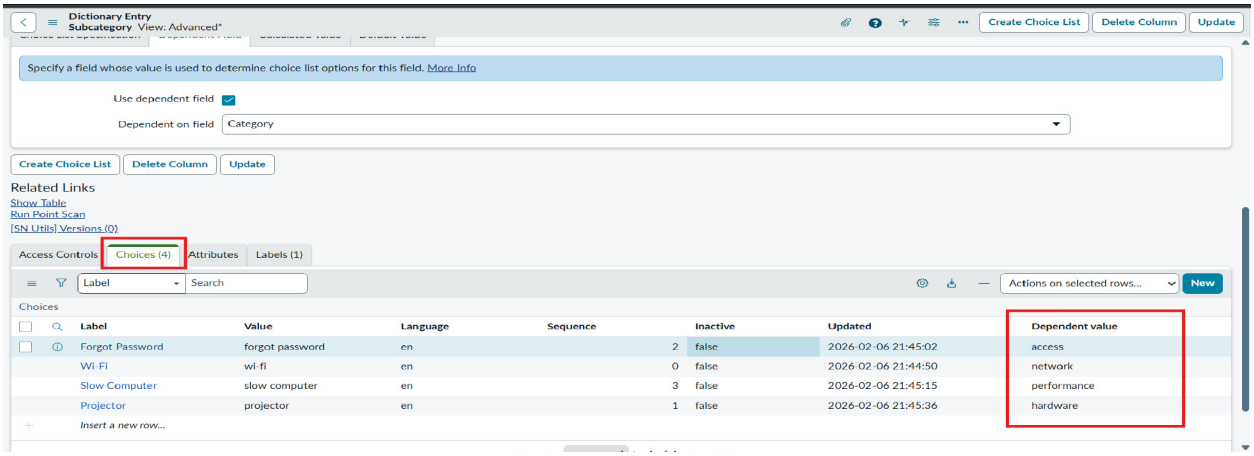
Navigation Steps

- Open the form of the table you created and right-click the **Subcategory** field.
- Select **Configure Dictionary**.
- Under Related Links, switch to **Advanced view**.
- Go to the **Dependent Field** tab.
- Tick **Use dependent field**.
- Choose **Category** in the **Dependent on field** box.
- Save the form.

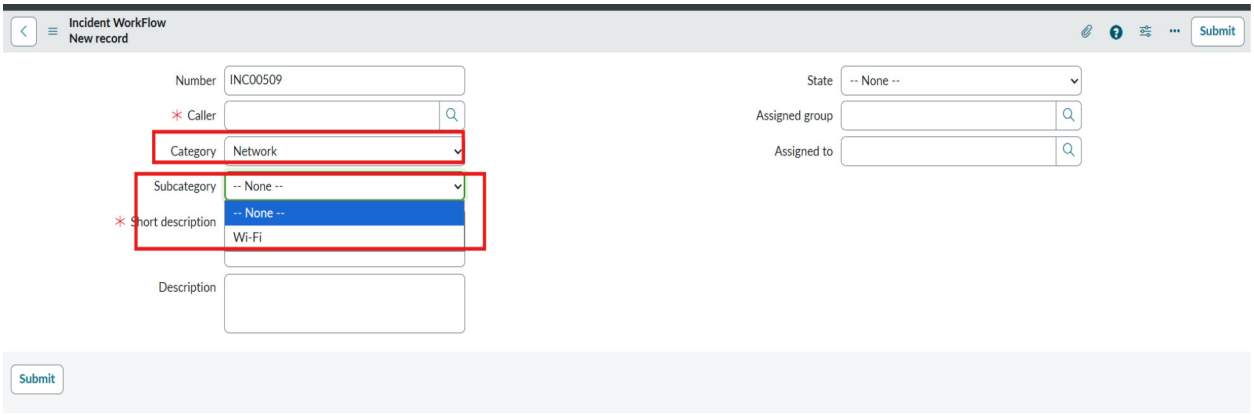


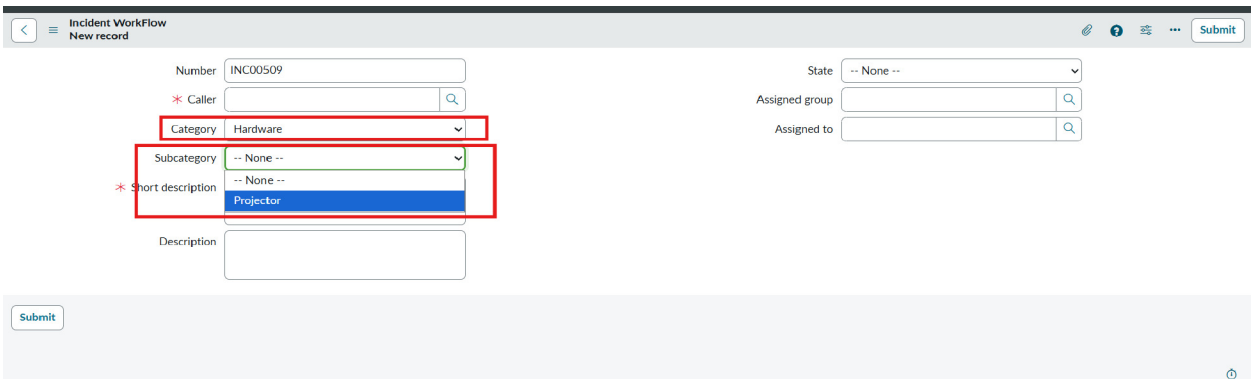


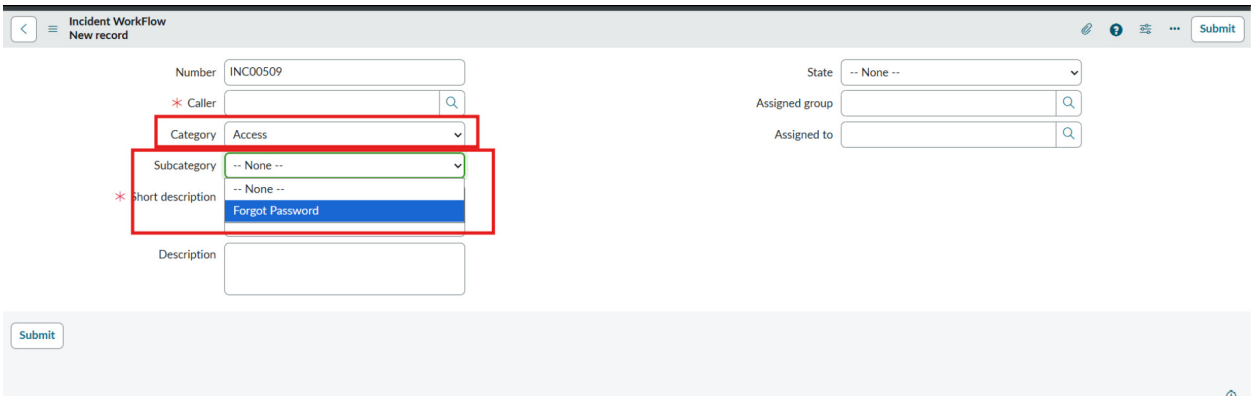
- Scroll down to the Choices section.
- For each subcategory choice, enter the backend value of its category as the dependent value:
- Wi-Fi → Network
- Projector → Hardware
- Forgot Password → Access
- Slow Computer → Performance

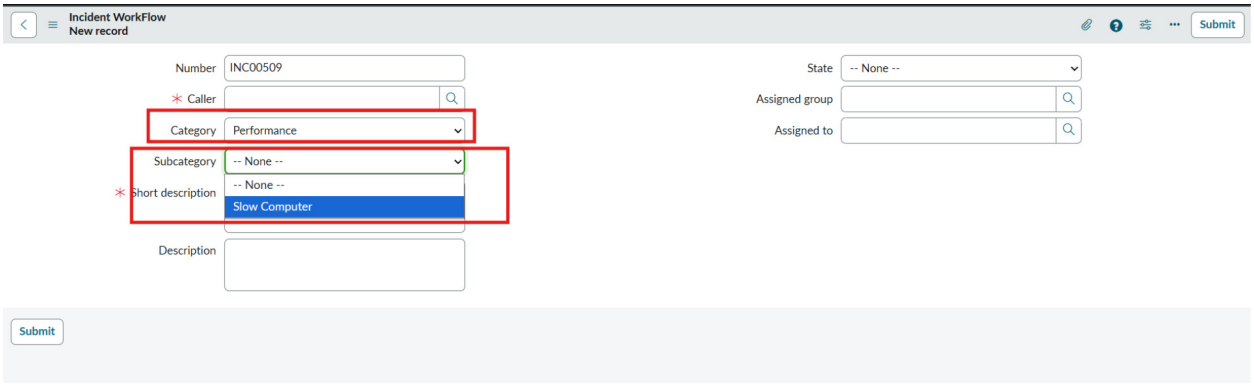


- Save the form.
- Each Category now displays only the Subcategory choice that belongs to it.


Incident WorkFlow
New record
Submit
Number
INC00509
Caller
Category
Network
Subcategory
-- None --
Short description
-- None --
Wi-Fi
Description
State
-- None --
Assigned group
Assigned to
Submit


Incident WorkFlow
New record
Submit
Number
INC00509
Caller
Category
Hardware
Subcategory
-- None --
Short description
-- None --
Projector
Description
State
-- None --
Assigned group
Assigned to
Submit


Incident WorkFlow
New record
Submit
Number
INC00509
Caller
Category
Access
Subcategory
-- None --
Short description
-- None --
Forgot Password
Description
State
-- None --
Assigned group
Assigned to
Submit


Incident WorkFlow
New record
Submit
Number
INC00509
Caller
Category
Performance
Subcategory
-- None --
Short description
-- None --
Slow Computer
Description
State
-- None --
Assigned group
Assigned to
Submit

PHASE 3: Automation with Flow Designer & Email Notification

When a student raises a new IT ticket, the only things they fill in are the Caller and a Short Description of the problem. They never have to choose a Category or Subcategory. The moment the ticket is submitted, a Flow Designer flow starts, checks the Short Description for preset keywords such as Wi-Fi, Projector, Password or Slow, and then sets the matching Category and Subcategory on the ticket.

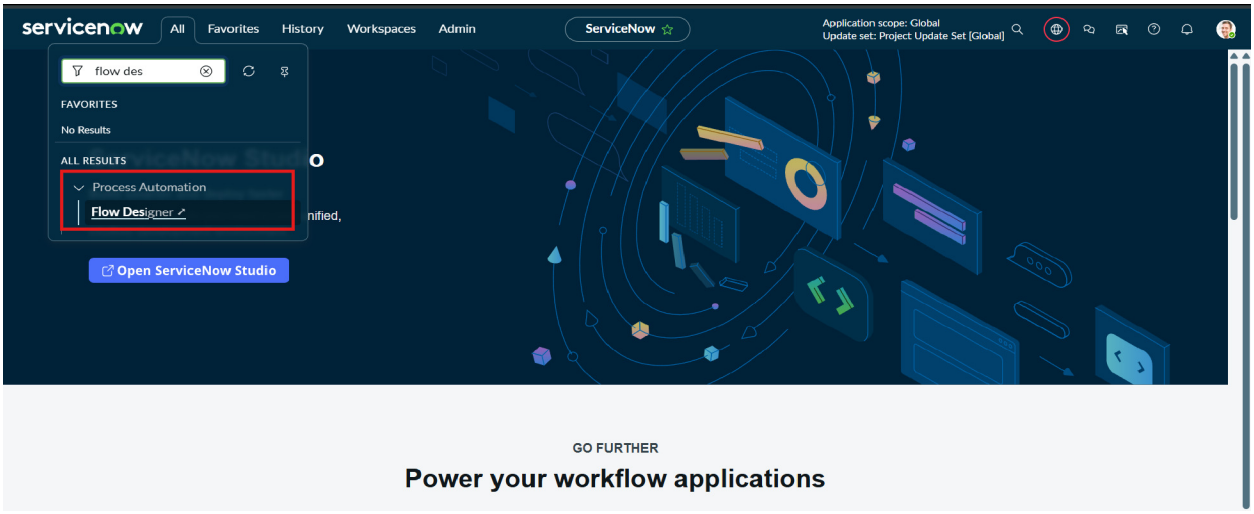
- Short description contains **Wi-Fi**: Category = Network, Subcategory = Wi-Fi.
- Short description contains **Projector**: Category = Hardware, Subcategory = Projector.
- Short description contains **Password** or **Login**: Category = Access, Subcategory = Forgot Password.
- Short description contains **Slow** or **Hanging**: Category = Performance, Subcategory = Slow Computer.

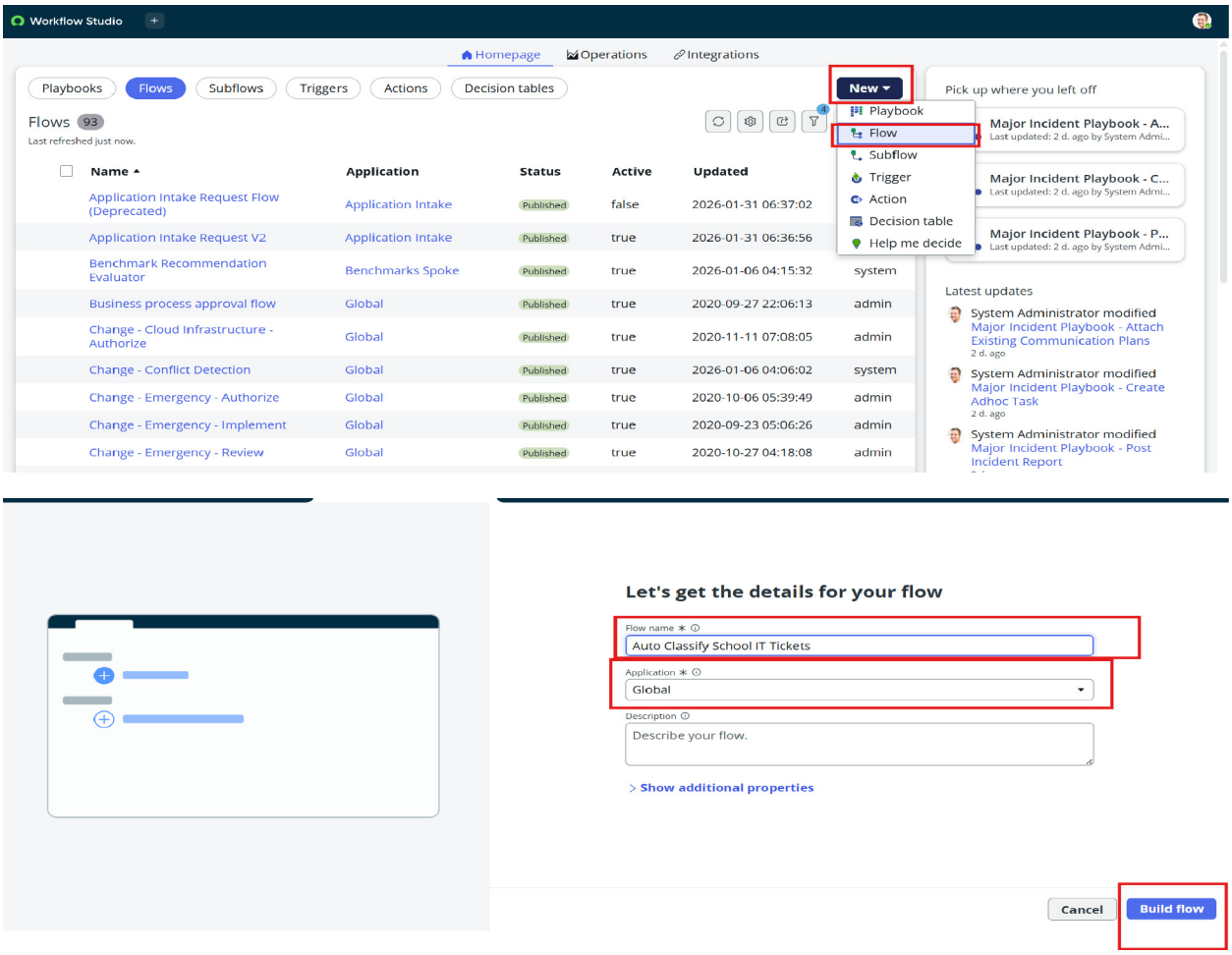
Since the same rules are applied every time, tickets are classified consistently, human error is removed and IT staff save a great deal of effort.

Everything is built inside Flow Designer, so the solution is easy to maintain, simple to scale and completely no-code.

Activity 1: Building the Auto-Classification Flow

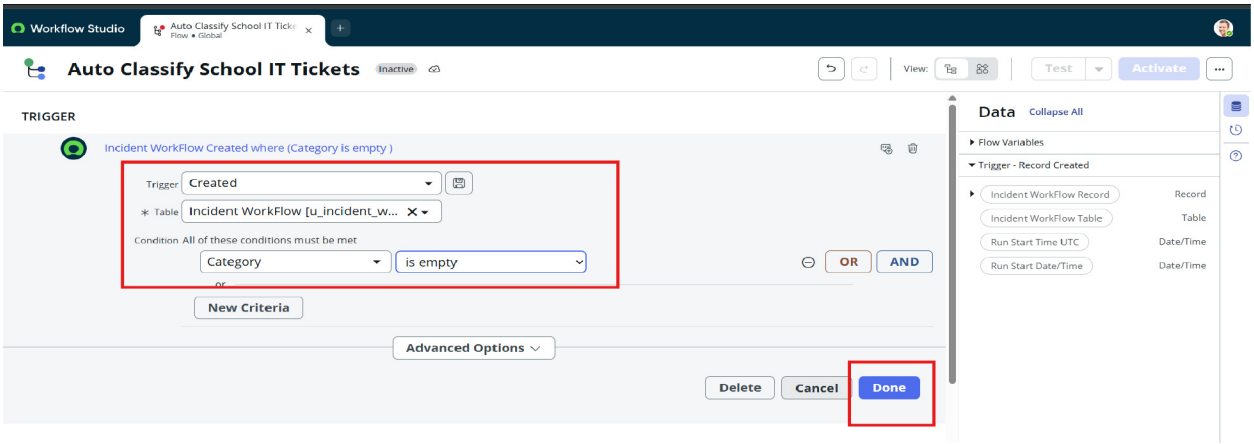
- Open the All menu and search for Flow Designer.
- Under Process Automation, open **Flow Designer**.
- Click **New** at the top right and choose **Flow**.
- Name the flow **Auto Classify School IT Tickets**.
- Set Application to **Global**.
- Click **Build Flow**.





Activity 2: Configuring the Trigger

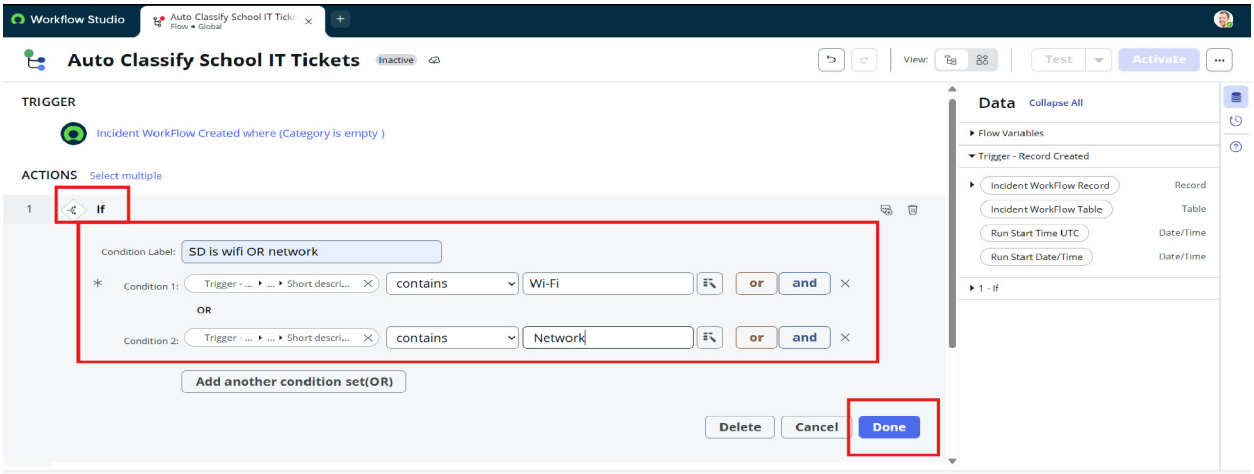
- Trigger: **Record Created**.
- Table: **Incident Workflow**.
- Condition: **Category is empty**.
- Click **Done**.



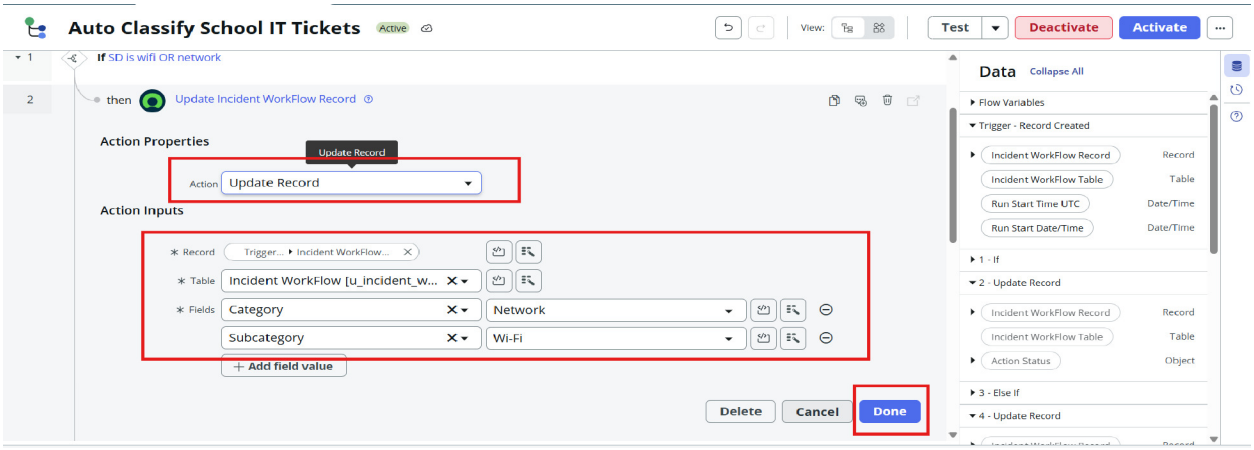
Next, under Actions, choose the **Flow Logic** option.

Activity 3: Wi-Fi Issue Classification

- Add an **If** condition.
- Give it the label **SD is Wifi or Network** (any label of your choice will do).
- Condition 1: use the data pill picker to select **Short description**, set the operator to **Contains** and the value to **Wi-Fi**.
- Condition 2: pick **Short description** again, operator **Contains**, value **Network**.
- Click **Done**.

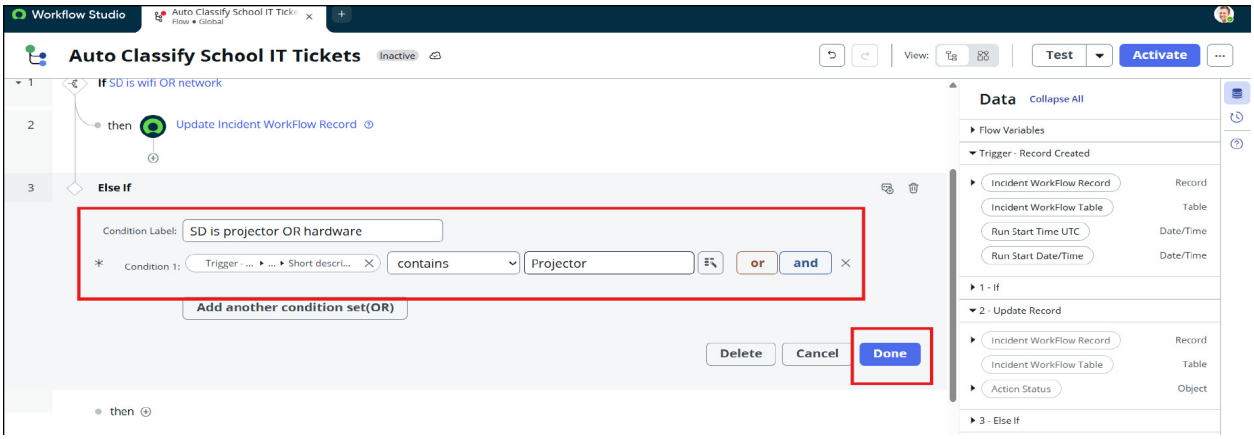


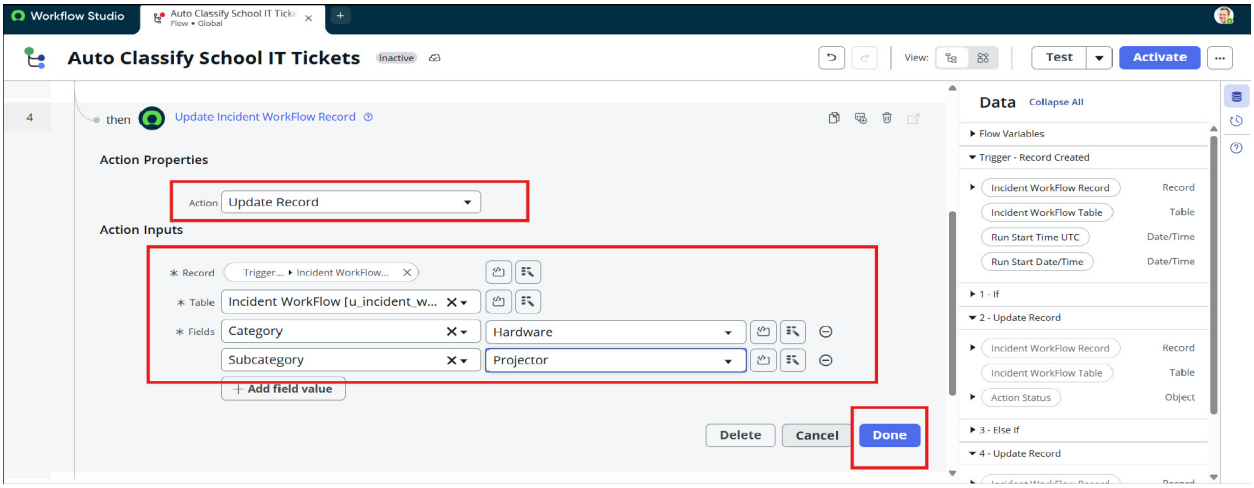
- Now add an **Update Record** action.
- Under Action Inputs, drag the record from the data panel on the right into the Record field.
- The Table fills in by itself as Incident Workflow.
- Set **Category = Network**.
- Set **Subcategory = Wi-Fi**.
- Click **Done**.



Activity 4: Projector Issue Classification

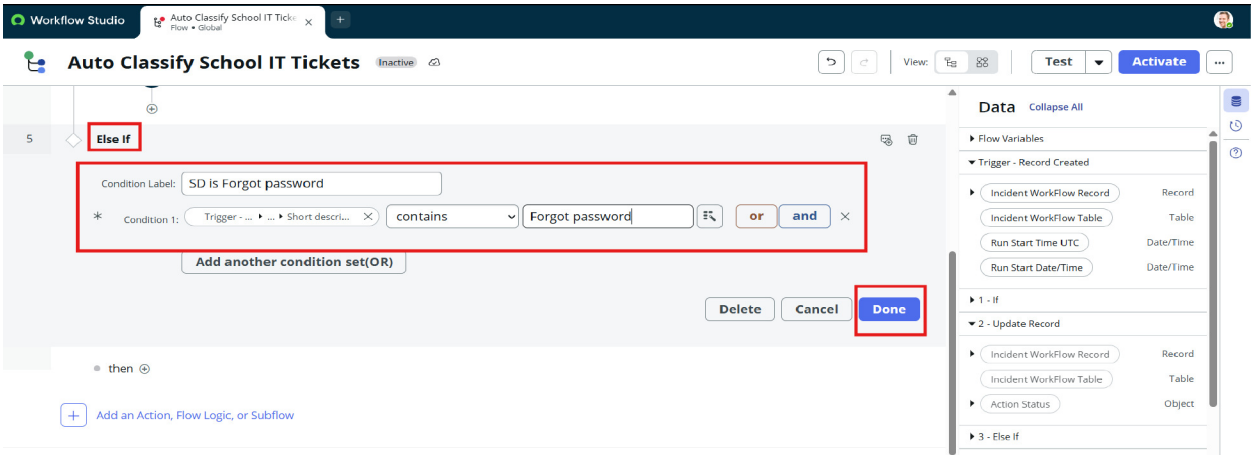
- Add an **Else If** condition.
- Label it **SD is Projector or Hardware**.
- Condition: Short description **Contains Projector**.
- Click **Done**.
- Add an **Update Record** action.
- Set **Category = Hardware**.
- Set **Subcategory = Projector**.
- Click **Done**.

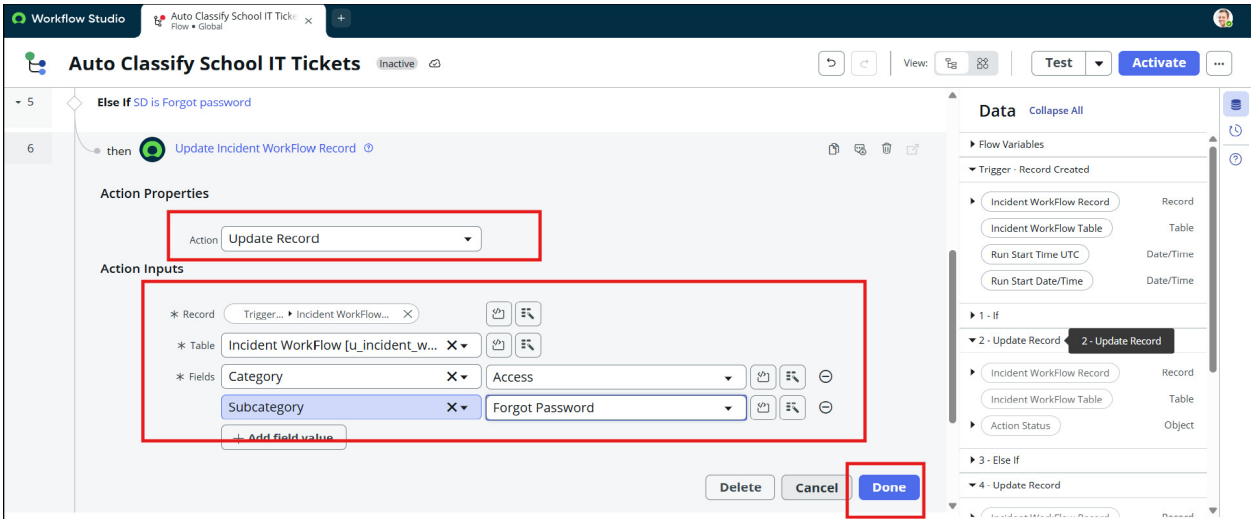




Activity 5: Password Issue Classification

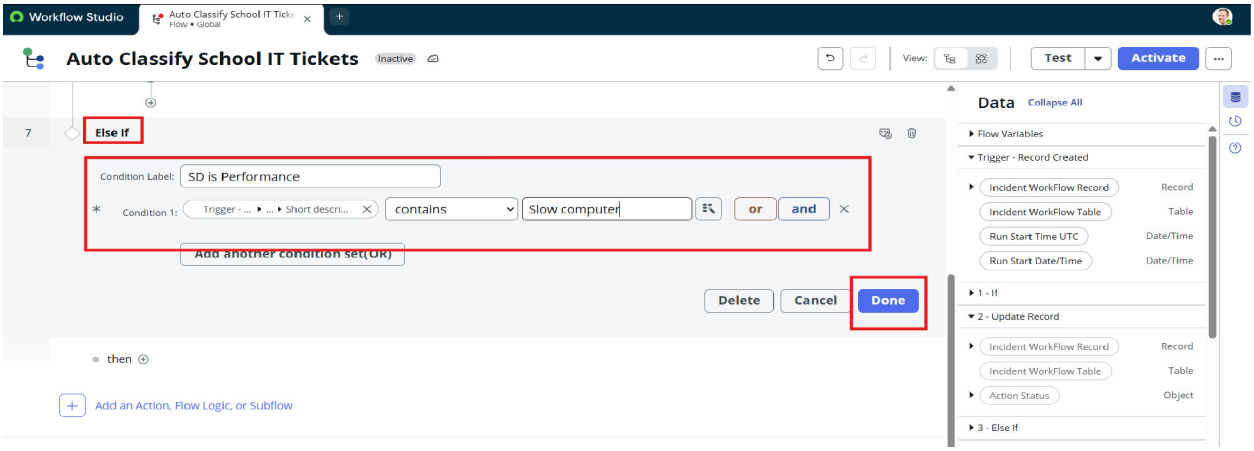
- Add another **Else If** condition.
- Label it **SD is Forgot password**.
- Condition: Short description **contains Forgot password**.
- Click **Done**.
- Add an **Update Record** action.
- Set **Category = Access**.
- Set **Subcategory = Forgot Password**.
- Click **Done**.

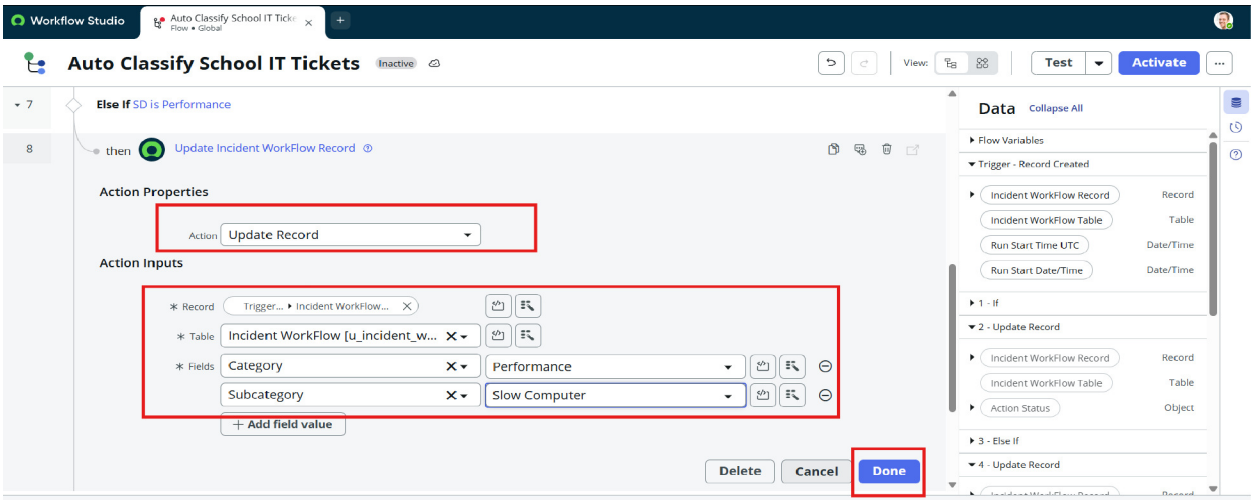




Activity 6: Slow Computer Classification

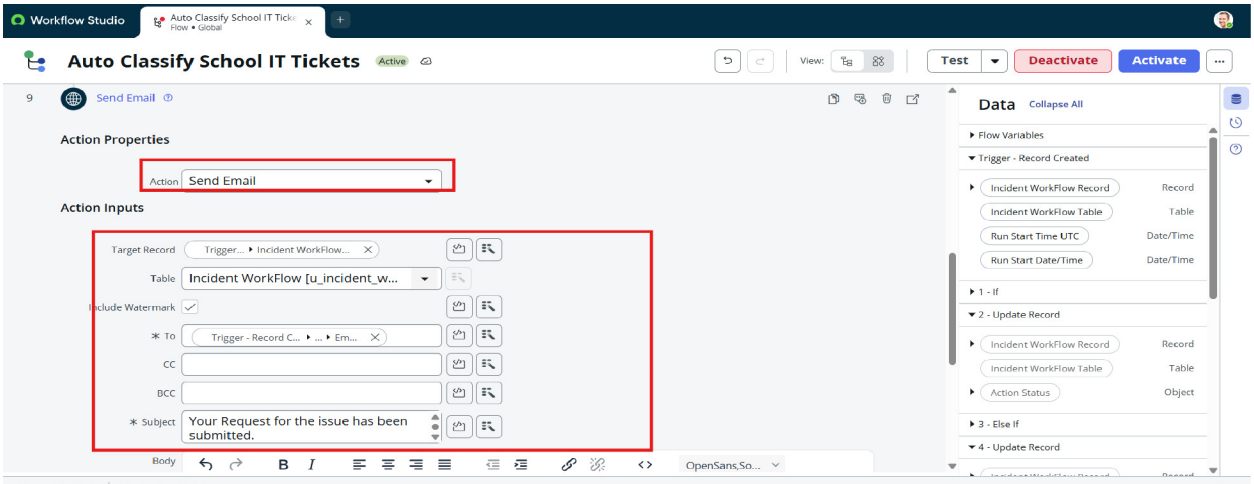
- Add one more **Else If** condition.
- Label it **SD is Performance**.
- Condition: Short description **contains Slow Computer**.
- Click **Done**.
- Add an **Update Record** action.
- Set **Category = Performance**.
- Set **Subcategory = Slow Computer**.
- Click **Done**.

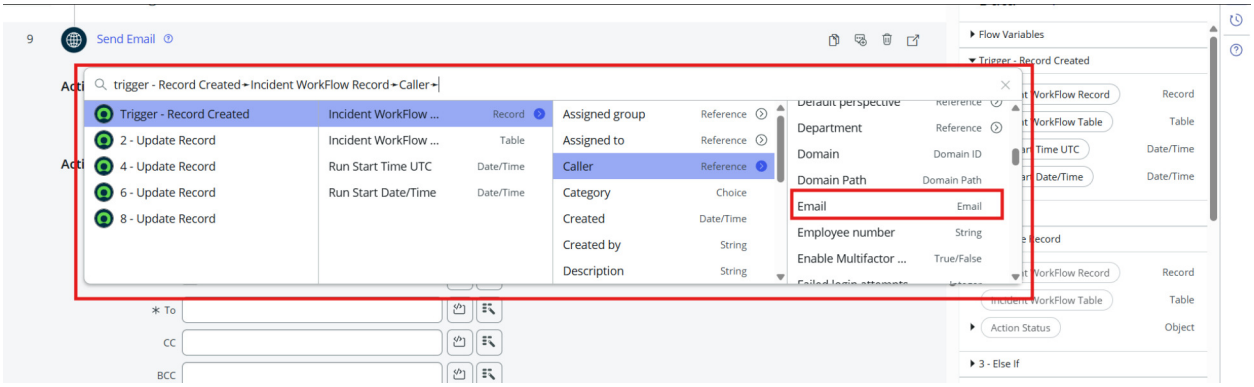




Activity 7: Setting Up the Email Action

- Once classification is done, the flow sends an email notification.
- Choose the action **Send Email**.
- To: the **Caller's Email**.
- Subject: **Your Request for the issue has been submitted**.
- Body: a ticket confirmation message.
- Click **Done**.





Finally, click **Activate** at the top right of the screen.

PHASE 4: Testing, Validation & Security

- This phase confirms that the finished solution behaves as intended, meets every functional requirement and follows basic security and access-control good practice.
- It checks the automation from start to finish, verifies that the data is accurate, and makes sure that only authorised users can view or change ticket data.

Data Validation Checks

- Mandatory fields are captured properly.
- Auto numbers are generated with no duplicates.
- Category and Subcategory values are stored correctly.
- Reference fields resolve to the right user and group records.

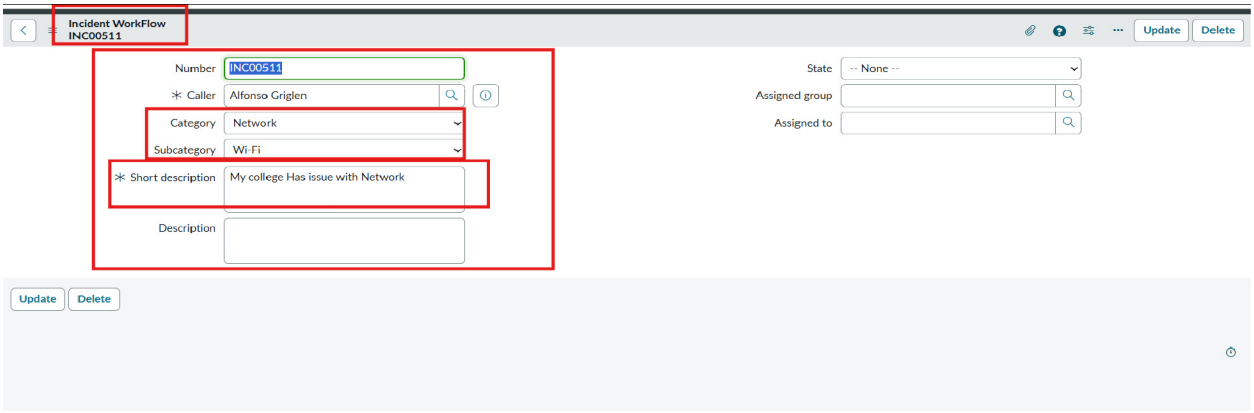
Activity 1: Testing and Validation

Test Scenario 1

- Select a caller.
- Enter a short description about a Wi-Fi or network problem, for example “WiFi not working in library”.
- Save the form.
- Reload the form.

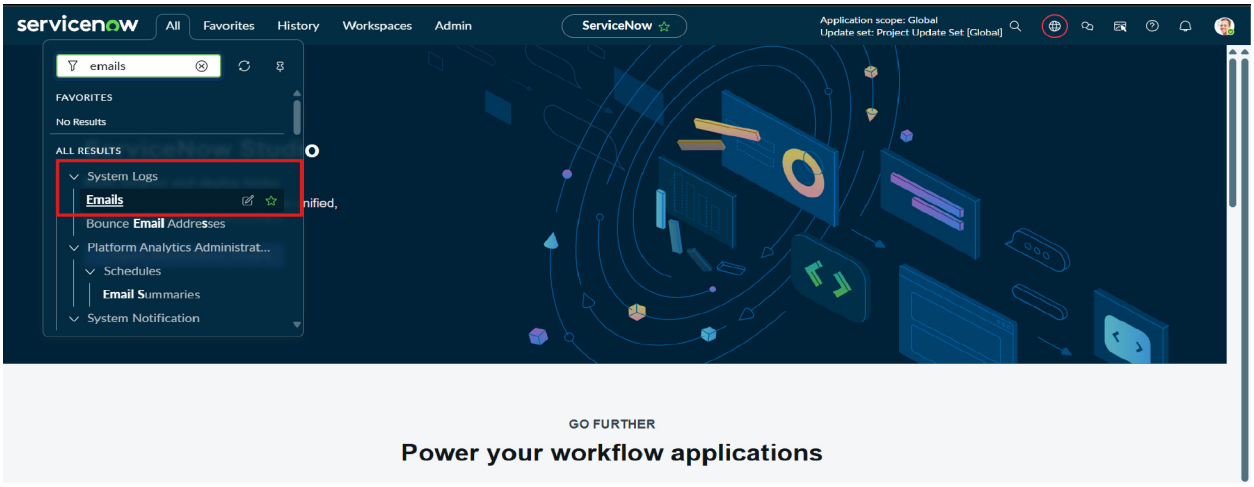
The fields are then filled in automatically:

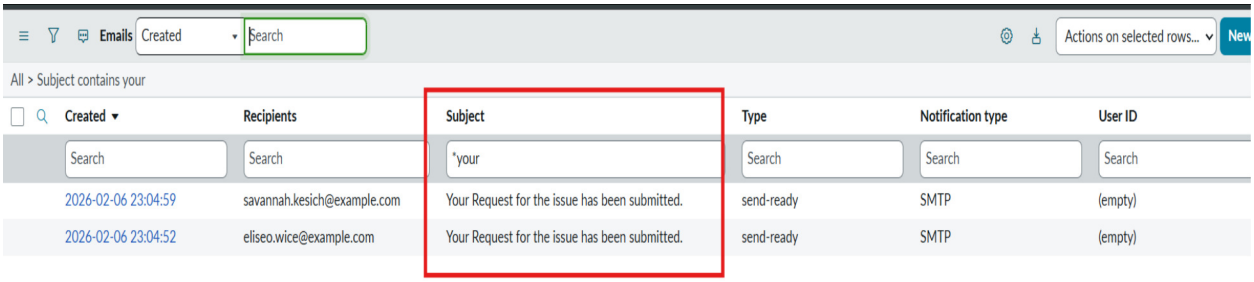
- Category → **Network**.
- Subcategory → **Wi-Fi**.
- An email is sent to the caller’s email address.

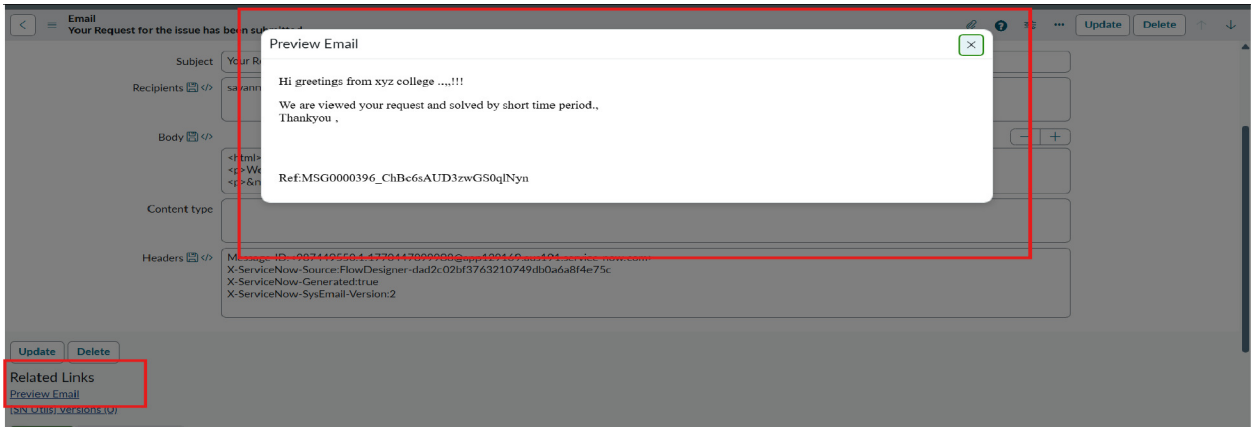


Activity 2: Checking the Email Notification

- Open the All menu and search for Emails.
- Under System Logs, select **Emails**.
- Search using the subject line given in the flow.
- Open the record and choose **Preview Email** under Related Links.





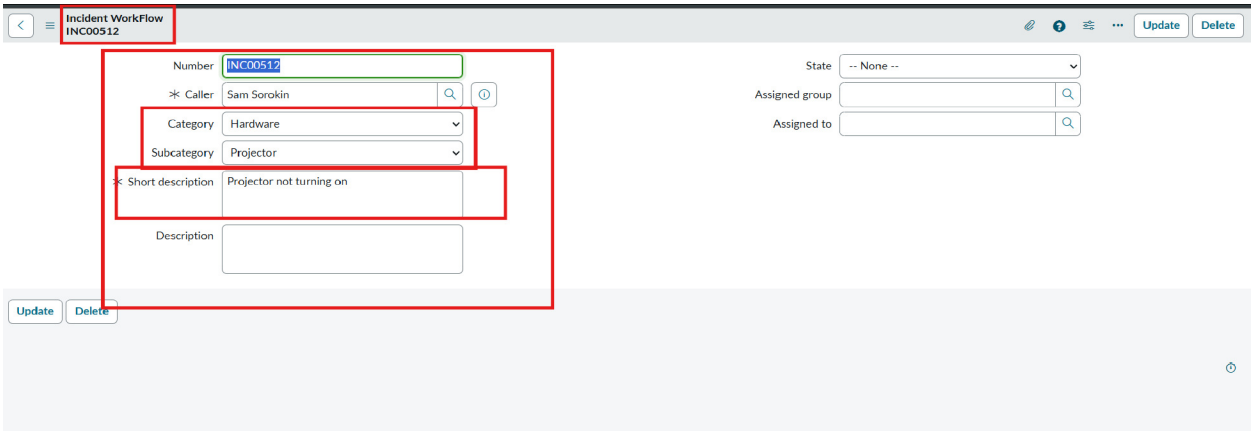


Activity 3: Test Scenario 2

- Select a caller.
- Enter the short description “Projector not turning on”.
- Save the form.
- Reload the form.

The fields are then filled in automatically:

- Category → **Hardware**.
- Subcategory → **Projector**.
- An email is sent to the caller’s email address.

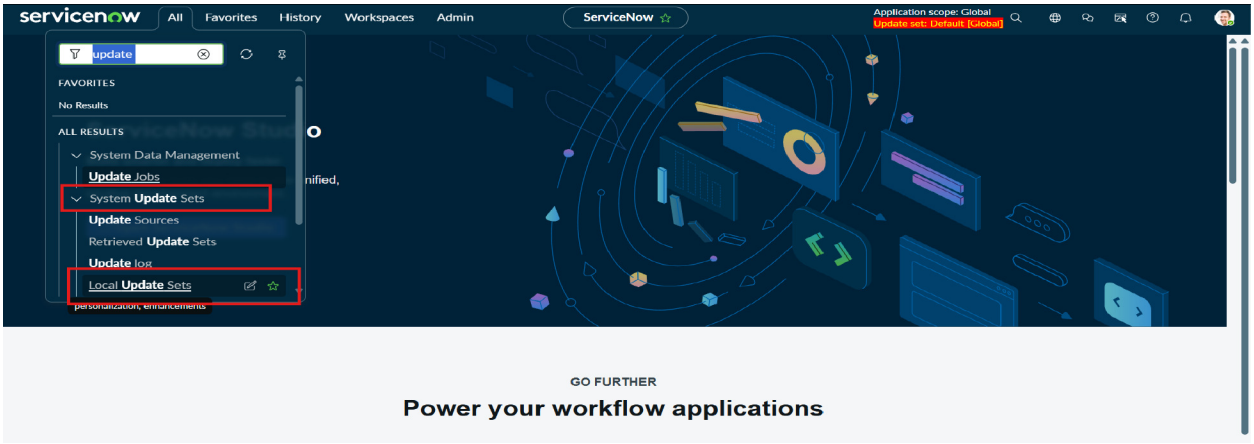


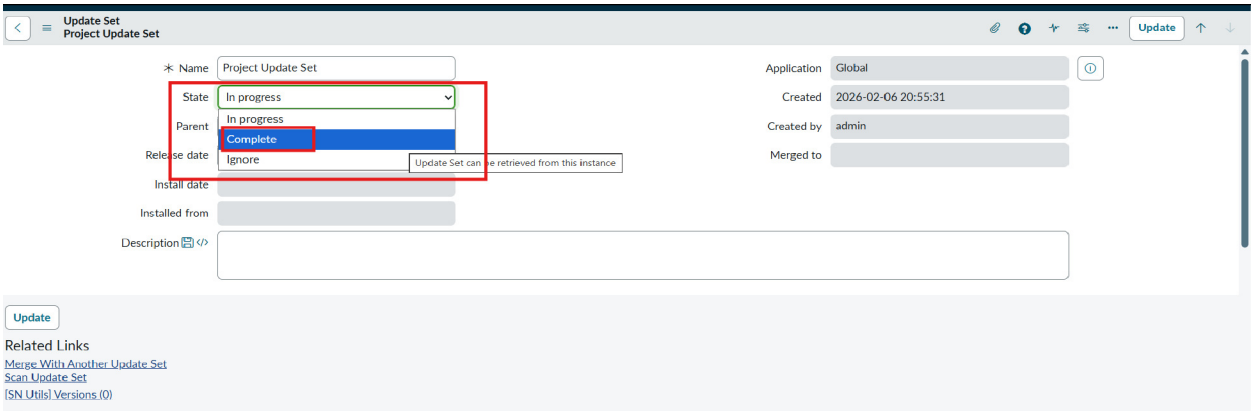
PHASE 5: Deployment & Final Documentation

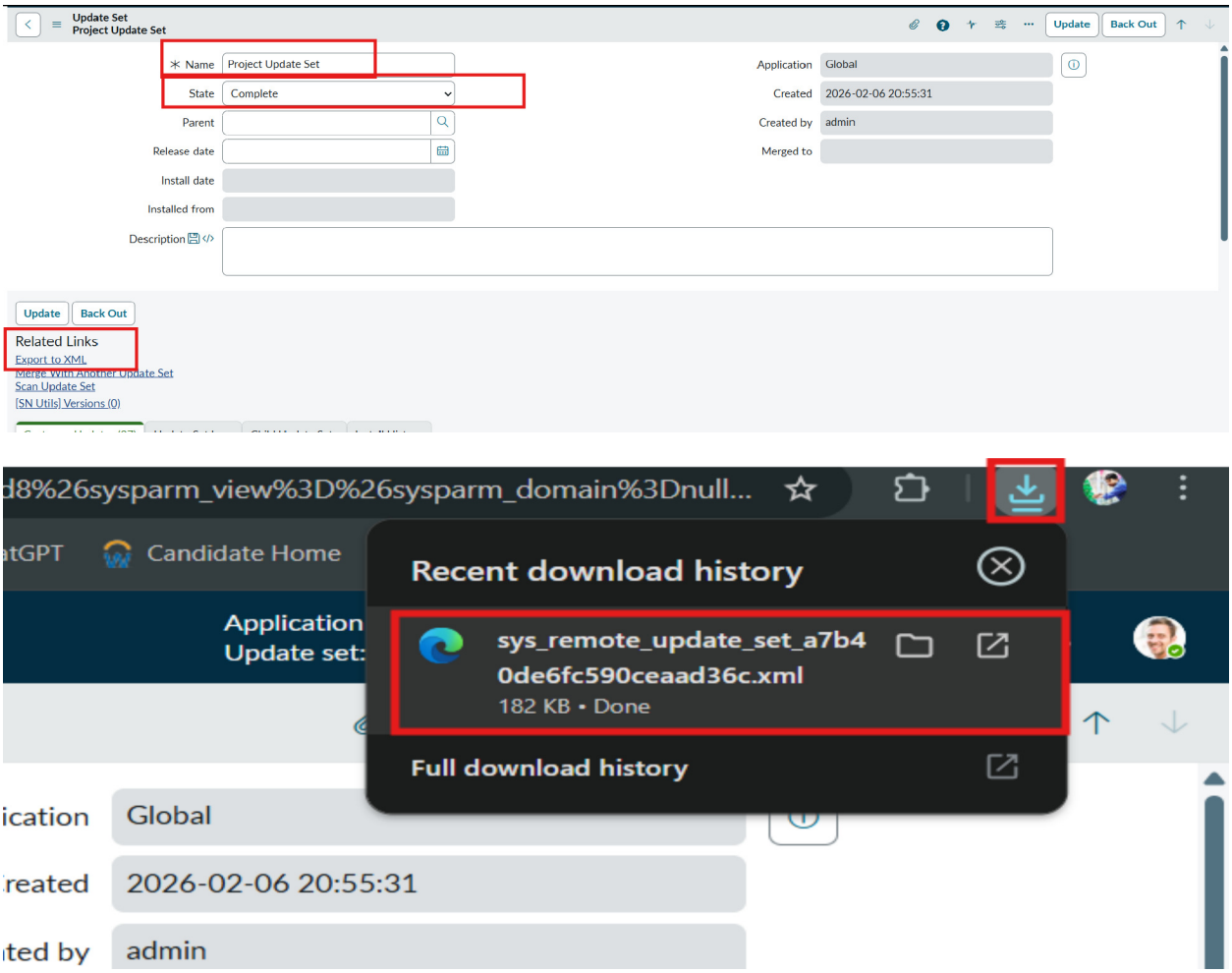
- This phase moves the finished solution to the target environment and completes all of the project documentation.
- It leaves the solution stable, reusable and ready to be demonstrated or used in production, backed by clear and complete documentation for later reference.

Activity 1: Moving the Update Set to Complete

- Open the All menu and search for Local Update Sets.
- Under System Update Sets, choose **Local Update Sets**.
- Open **Project Update Set**.
- Change the State from **In progress** to **Complete**.
- Save the form.
- In Related Links, click **Export to XML**.
- The update set is downloaded to your computer as an XML file.
- Send that file to others through WhatsApp or email.
- This is how the update set can be carried over to another instance.







Conclusion

- The Auto Ticket Classification with Flow Designer project provides a complete, end-to-end automation for a school IT helpdesk. Its main aim, cutting the manual effort of sorting tickets and speeding up responses, has been fully met.
- By combining careful requirement analysis, solid backend configuration and no-code automation in Flow Designer, the system now sorts IT tickets by itself using the short description that students enter. Dependent choice fields keep the data accurate, and automatic emails let callers know immediately that their ticket has been received.
- A phased approach kept the build clear, stable and easy to maintain. Testing and security checks showed that the system works reliably, keeps its data intact and enforces suitable access controls.
- Overall, the project shows how ServiceNow can deliver scalable, maintainable and user-friendly automation without complex scripting or machine learning. It suits small and medium settings such as schools and colleges, and can later be extended with features like automatic assignment, SLA tracking or predictive intelligence.